AAA COLLEGE OF ENGINEERING AND TECHNOLOGY

DECENTRALIZED VOTING SYSTEM USING ETHEREUM BLOCKCHAIN

Presented By

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Abstract

- Traditional voting systems face fraud and trust issues. This project uses blockchain and smart contracts to create a secure, transparent, and tamper-proof voting system. Voter privacy is maintained while ensuring accurate, real-time results. The system is built with Ethereum, Solidity, and Python, and can be scaled for larger elections.

Objective

- Provide a secure and transparent voting system.
- Prevent vote tampering and fraud.
- Ensure voter privacy and anonymity.
- Enable real-time vote counting and result verification.
- Reduce dependency on centralized authorities.
- Allow salability for different election sizes.
- Improve trust and efficiency in the voting process.
- Explore future enhancements like advanced encryption techniques.

Suggestions/Corrections Given By Review Committee

Problem Definition

✓ **Fraud and Manipulation**

- Use an **immutable blockchain ledger** so votes cannot be altered, forged, or deleted.
- Implement **smart contracts** to allow only one vote per registered voter, preventing multiple or unauthorized voting.

✓ **Lack of Transparency**

- Allow **public verification** by providing audit trails through the blockchain explorer.
- Use **automated vote counting** to ensure accurate and verifiable results without human intervention.

✓ **Delays in Counting and Result Declaration**

- Provide **real-time vote counting** as votes are cast, eliminating processing delays.
- Automatically **publish results** once voting ends, reducing time between election closure and outcome declaration.

✓ **Privacy Concerns**

- Apply **cryptographic encryption** to protect votes and ensure sensitive data is secure.
- Authenticate voters without revealing identities, enabling **anonymous voting** with secure access controls.

Introduction

- Voting is essential for democracy and decision-making.
- Current systems face issues like fraud, delays, and lack of trust.
- Blockchain technology offers a secure and transparent way to conduct elections.
- This project aims to build a decentralized voting system using smart contracts and blockchain to ensure fairness, privacy, and efficiency.

Literature Survey

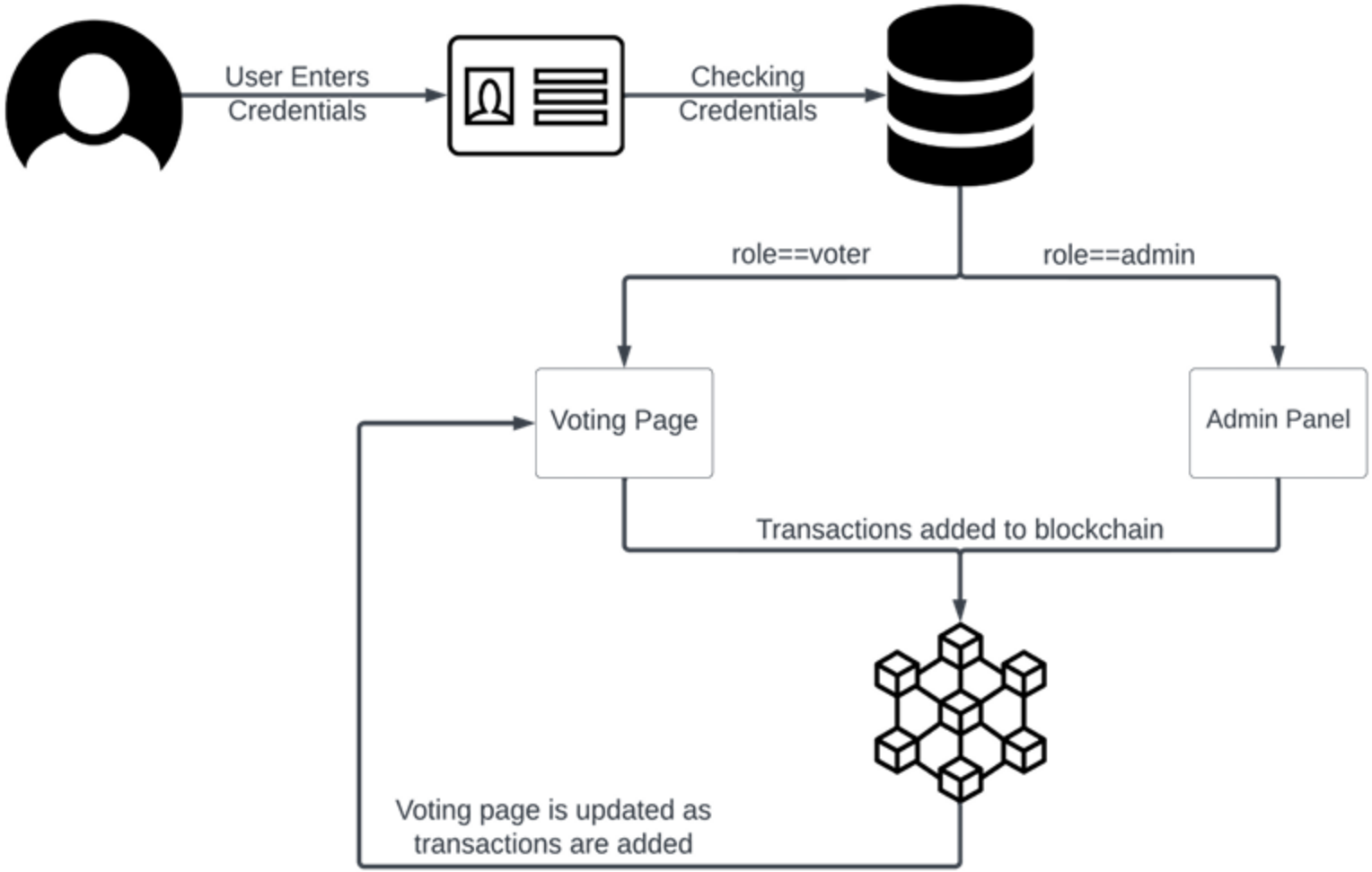
S.NO	TITLE	AUTHOR	YEAR	METHODOLOGY	PROBLEM IDENTIFIED
1	A Framework to make Voting System Transparent Using Blockchain Technology	M.S.Farooq,U .Iftikhar, A.Khelifi	2022	Developed a blockchain framework using consensus protocols to ensure immutability & transparency.	Traditional voting lacks trust, transparency, and tamper-resistance.
2	E-Voting System Using Blockchain Technology	Nagesh H.R, Guru Prasad M.S, Shivaraj B.G., Et al.	2022	Implemented Ethereum smart contracts with Ganache/Remix for prototype.	Centralized online voting vulnerable to fraud and manipulation.
3	BieVote: Biometric-enabled Blockchain Voting Framework	M.J.H.Faruk,H .Shahriar, et al.	2022	Integrated biometric authentication with blockchain-based voting.	Weak voter authentication in existing e-voting systems.

S.NO	TITLE	AUTHOR	YEAR	METHODOLOGY	PROBLEM IDENTIFIED
4	**Secure Framework for E-Voting Using Blockchain**	**A. Author et al.**	**2022**	**Designed smart contract-based secure voting architecture.**	**Risk of data tampering and lack of secure storage.**
5	Blockchain-enabled Decentralized Trust Management & Secure Voting	R. S. Reddaiah, R. Subramanyam, et al.	2024	Proposed decentralized trust model with blockchain primitives.	Lack of trust management and vulnerability to manipulation in centralized systems
6	BP-Vot: Blockchain-based e-Voting using Smart Contracts, Differential Privacy & SSI	H. Baniata, G. Caluna	2025	Used smart contracts with differential privacy & self-sovereign identity for privacy-preserving voting.	Existing blockchain voting lacks strong privacy & scalability for large elections.

Proposed Method

- Use Ethereum blockchain to build a decentralized voting system.
- Provide a transparent and tamper-proof election process.
- Leverage smart contracts for secure and anonymous voting.
- Ensure integrity and immutability of voting data.
- Increase voter trust and confidence in the process.
- Reduce risks of fraud and manipulation.

Architecture Diagram



Hardware and Software Requirements

Hardware	Software
Processor : Intel Core i3 or above RAM: 4GB Or above Storage:250GB Internet : Stable broadband connection Optional : Webcam or Fingerprint scanner for demo	OS : Windows 10 or 11 Language : Python Blockchain Tools : Ganache, Meta mask. Smart Contracts: Solidity Frontend :HTML, CSS, JavaScript Database : MySQL IDE:VS Code

Modules

- Voter Registration
- Verify voter
- Voting
- Vote Counting
- Result Declaration
- Security
- Candidate Registration

Modules Explanation

- **Voter Registration** – Registers and verifies eligible voters.
- **Voting** – Casts votes securely and anonymously.
- **Vote Counting** – Counts votes automatically and accurately.
- **Result Declaration** – Displays verified results.
- **Security** – Prevents fraud and protects voter privacy.
- **Candidate Registration** – Adds candidates on the blockchain.

Methodology

Step 1 – System Initialization:

- Ethereum test network (e.g., Ganache) is configured.
- Smart contract is deployed to manage elections.

Step 2 – Voter and Candidate Registration:

- Admin registers candidates.
- Eligible voters are verified and assigned blockchain addresses.

Step 3 – Voting Phase:

- Each voter logs in using a secure ID.
- Vote is cast through the smart contract; transaction recorded on blockchain.

Step 4 – Vote Counting:

- Blockchain ensures all votes are immutable and counted automatically.
- Smart contract prevents double voting.

Step 5 – Result Declaration:

- Smart contract tallies and publishes results in real time.
- Results can be verified by anyone on the blockchain explorer.

Algorithm

1. Admin deploys election smart contract.
2. Register candidates → Store candidate IDs in blockchain.
3. Register voters → Assign unique blockchain address.
4. If voter.hasVoted == false, allow vote; else reject.
5. Voter submits vote → Transaction recorded on Ethereum blockchain.
6. Smart contract increments vote count of selected candidate.
7. When voting ends, call getResults() function.
8. Display results securely via DApp interface.

Expected Output

Candidate Registered: Candidate_A

Voter Registered: Voter_1 (0xAb3...F29)

Vote Cast Successfully!

Transaction ID: 0x2f6e3a91d2a...

Winner: Candidate_A with 63% votes

References

1. M. S. Farooq, U. Iftikhar, and A. Khelifi, “A framework to make voting system transparent using blockchain technology,” *IEEE Access*, vol. 10, pp. 122233–122244, 2022.
2. Nagesh H. R., M. S. Guru Prasad, B. G. Shivaraj, et al., “E-voting system using blockchain technology,” in *Proc. IEEE Int. Conf. on Advances in Computing, Communication and Control (ICAC3N)*, 2022.
3. M. J. H. Faruk, H. Shahriar, et al., “Bie Vote: A biometric identification-enabled blockchain-based secure and transparent voting framework,” in *Proc. IEEE Int. Conf. on Blockchain and Cryptocurrency (BCCA)*, 2022.
4. A. Author et al., “A secure framework for e-voting using blockchain,” in *Proc. IEEE Int. Conf. on Computer Science, Engineering and Applications (ICCSEA)*, 2022.
5. R. S. Reddaiah, R. Subramanyam, et al., “Blockchain-enabled decentralized trust management and secure voting system,” in *Proc. IEEE Int. Conf. on Big Data and Smart Computing (ICBDS)*, 2024.
6. H. Baniata, G. Caluna, “BP-Vot: Blockchain-based e-voting using smart contracts, differential privacy, and self-sovereign identities,” *IEEE Access*, vol. 13, pp. 41234–41247, 2025.

Thank You